Class 4 Maths Olympiad — Prep Kit

A self-contained practice kit for Class 4 (age 9–10) Mathematics Olympiad exams — SOF-IMO, Silverzone iOM, Unified Council UIMO, and similar school-level olympiads. It covers the syllabus, the exact exam pattern, a week-by-week practice plan, topic worksheets, and full-length timed sample papers.

Primary target exam

The pattern and worksheets are built against **SOF International Mathematics Olympiad (IMO), Class 4** — the most widely taken olympiad of this type — because its 4-section structure (Logical Reasoning / Mathematical Reasoning / Everyday Mathematics / Achievers) is a superset of what Silverzone iOM and UIMO test. Topic overlap with those two is mapped out in `docs/01-syllabus.md`, so the same worksheets double as prep for any of the three.

Repository layout

```
docs/
  01-syllabus.md          Full topic-wise syllabus, section by section
  02-exam-pattern.md      Marks, timing, levels, negative marking, cutoffs
  03-practice-plan.md     12-week schedule: what to practice, and when
assignments/
  README.md               Naming convention + full topic list (Weeks 1–11 + optional)
  01-number-sense/ … 11-achievers-hots/   11 core topic worksheets, each with an answer key
  12-math-kangaroo-stretch/               Optional: Math Kangaroo-style puzzle stretch
  13-achievers-level2/ … 15-geometry-level2/   Level-2 topic worksheets (harder, for Round-1 qualifiers)
sample-papers/
  README.md               How the full-length papers map to the real pattern
  sample-paper-01.md … sample-paper-09.md     9 full 35-question papers (IMO pattern, 40 marks, 60 min)
  sample-paper-01-answer-key.md … sample-paper-09-answer-key.md
                          Papers 1–5: Level-1. Papers 6–9: Level-2 track.
pdf/
  Print-ready PDF of every file above, mirroring the same folder structure
tools/
  generate-pdfs.js        Regenerates pdf/ from the Markdown (see below)
```

How to use this

1. Read `docs/01-syllabus.md` and `docs/02-exam-pattern.md` once, so you know what "done" looks like.
2. Follow `docs/03-practice-plan.md` week by week — it tells you which assignment to do and when to attempt a timed sample paper.
3. Every assignment and sample paper is self-contained: questions first, answer key (with short explanations) in a separate file, so you can mark your own work.
4. Sample papers are meant to be done **strictly timed** (60 minutes, no calculator) to build real exam pace.

Regenerating PDFs

If you edit any `.md` file, its PDF under `pdf/` won't update itself — rebuild all of them with:

```
cd tools
npm install
```

```
npx playwright install chromium   # one-time, downloads a local Chromium
node generate-pdfs.js
```

It auto-discovers every `.md` file in the repo (title comes from each file's first `# heading`), so a new assignment or sample paper just needs to exist — no code changes required. It also resolves Markdown images (e.g. the SVG diagrams under `assignments/*/images/`) relative to the source file they're referenced from.

Status

Complete. Syllabus/pattern research, the 12-week plan, all 11 weekly assignments (Weeks 1–11), and all 9 full-length sample papers (5 Level-1 + a 4-paper Level-2 track for students who qualify Round 1) are built out — each with an answer key, and each arithmetic/logic answer verified programmatically. PDF versions of every file are in `pdf/`. An optional 15-question Math Kangaroo-style stretch set (`assignments/12-math-kangaroo-stretch/`) is also built, for after the core plan is comfortably done. Three dedicated **Level-2 topic assignments** (Achievers/HOTS, Logical Reasoning, Geometry — the areas where Level 2 gets conceptually harder, not just numerically bigger) are also built, in `assignments/13-achievers-level2/` through `15-geometry-level2/`.